

AEGIS | Universal Log Pre-processing Framework

SIH Problem Statement 26156 · NTRO · Two-page architecture brief

1. Purpose and processing flow

AEGIS receives perimeter telemetry through TCP/UDP Syslog, optional TLS Syslog, file upload and authenticated HTTP. It commits the exact received bytes before parsing. Every record receives an event ID, SHA-256, source, transport, peer, framing and receipt time. This preserves the collector evidence; UDP loss before receipt remains outside the framework control.

DEVICE FEEDS	CAPTURE	PRESERVE	NORMALIZE	OPERATE
Firewall · router · IDS TCP / UDP / TLS / file / HTTP	Bounded frames size + queue limits	Immutable raw bytes SHA-256 + receipt metadata	Built-ins or signed mapping AEGIS fields + unmapped	Dashboard · graph · cases ECS / JSONL / CSV / HTTP

Parser adaptation and review

Supported decoders include Syslog wrappers, JSON/Suricata, XML, CSV, CEF, LEEF, Cisco ASA, pfSense, key/value and unfamiliar text. Fingerprints and Drain3 templates identify new structures. A local model can propose mappings, but deterministic type and meaning checks filter its output. A reviewer adds semantic assertions, validates replay vectors and approves a signed versioned plugin. Replay creates a new revision and leaves original evidence unchanged.

UNKNOWN / DRIFTED	DISCOVERY	VALIDATION	HUMAN GATE
Fingerprint + template Attributes retained	Drain3 + aliases Optional local model	WASM projection Replay + mutations Semantic assertions	Signed plugin Registry activation Replay retained events

2. Integrity, security and delivery

Approved field projection runs in Wasmtime with zero host imports, 64 KiB memory and 10,000 fuel. Model output is untrusted data, never executable code. Trusted public keys gate bundle imports. Raw and revision tables have append-only database triggers; the audit log is a signed hash chain. Evidence leaves are committed into Merkle batches. Three local Ed25519 witnesses sign ordered checkpoints with a 2-of-3 quorum. The default witnesses share one host, so the quorum demonstrates signed tamper evidence rather than independently administered custody.

CONTROL	WHAT AEGIS CHECKS	OPERATIONAL LIMIT
Evidence	Raw hash, manifest, Merkle inclusion, ledger signatures	Cannot prove upstream truth
Parser safety	WASM fuel/memory/import bounds; signed bundle and vectors	Trusted Python performs structural decoding
Delivery	Durable HTTP outbox, backoff, stable Idempotency-Key	At-least-once; receiver deduplicates
Drift	Source fingerprint baseline and review queue	Approved correction is required
Access	Bearer admin token, local assets, private host allowlist	Single admin role in prototype

3. Deployment and evaluation boundary

The prototype runs as a single SQLite-WAL node and ships a hardened non-root Docker configuration, local wheelhouse preparation, a controlled socket lab and a browser dashboard. The measured local benchmark is in docs/benchmark.json. The implementation does not claim a distributed Kafka/Flink/object-store cluster, independent remote witnesses, full OCSF certification, native Splunk/Elastic/Kafka/S3 connectors, trained anomaly detection or sustained billion-event throughput. Production scale requires partitioned storage, durable brokers, horizontal workers, WORM retention and independent witness services.

SIH REQUIREMENT	PROTOTYPE EVIDENCE
Raw lossless representation	Exact raw bytes + hash; JSONL export; immutable evidence tests
Plug-and-play onboarding	Fingerprint → candidate → semantic review → signed parser → replay
SIEM / data lake readiness	AEGIS envelope, ECS projection, JSONL/CSV, retrying HTTP outbox
Air gap	Local UI/API, pinned offline wheelhouse, local model adapter, no cloud fallback
Platform independence	Container files included; image execution needs a Docker daemon

Prepared for SIH evaluation · AEGIS 1.0 demo · Source and test evidence are included with the submission package.